KS3 DIGITAL TECHNOLOGY

ENGINEERING BIOLOGY AND HEALTH TECH

Where computing meets life science: how wearables measure your body, how AI helps doctors, how DNA is stored and read as data, how engineering biology and gene editing work, why health data needs protecting, and how to design a health-tech solution.

LESSONS

1 Wearables and the measured body
2 AI in medicine
3 DNA is data
4 Engineering biology
5 Your health data
6 Design a health-tech solution

LESSON 1 OF 6

WEARABLES AND THE MEASURED BODY

Wearables such as smartwatches and fitness trackers use sensors including accelerometers, optical heart-rate sensors and GPS to measure movement, heart rate, sleep and more.

ENGINEERING BIOLOGY AND HEALTH TECH

STARTER · 5 MINUTES

DO NOW

How many steps do you think you take in a school day? How could a device count them?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how wearable devices use sensors to measure the body, and evaluate the data they produce

SUCCESS CRITERIA: I CAN...

- ✓ I can describe the sensors in a wearable and what they measure
- ✓ I can explain how an optical heart-rate sensor works
- ✓ I can evaluate the accuracy and usefulness of wearable data

WEARABLES

- Smartwatches, fitness trackers, smart rings, sensor patches
- Measure steps, heart rate, sleep, location and more
- Data goes to an app, often via Bluetooth and the cloud
- Some are medical devices, e.g. continuous glucose monitors

KEY WORDS

WEARABLE	A small electronic device worn on the body that collects data with sensors.
ACCELEROMETER	A sensor that detects movement and changes in speed.
OPTICAL HEART-RATE SENSOR	A sensor that shines light into the skin to measure heart rate.
ACCURACY	How close a measurement or prediction is to the true value.
DIGITAL HEALTH	Using technology to support health and care.

THE SENSORS INSIDE

- Accelerometer: movement, used to count steps and detect falls
- Optical heart-rate sensor: green light shines into the skin
- Blood absorbs the light; each heartbeat changes how much is reflected
- GPS: distance and route; temperature sensors; some measure blood oxygen

CAN YOU TRUST THE NUMBERS?

- Good for trends over time, less reliable for single readings
- Errors from loose straps, skin tone, tattoos, movement
- Not a diagnosis: a doctor must interpret medical concerns
- Wellbeing: numbers can motivate, but can also cause worry

ACTIVITY · 20 MINUTES · PAIRS

DATA DETECTIVES

1. Look at the two weeks of (made-up) fitness tracker data for "Sam": steps, heart rate and sleep.
2. Describe two trends in the data.
3. Find one reading that looks wrong and suggest why.
4. Decide one helpful thing the data tells Sam and one thing it can't tell them.

In your workbook, answer:

? Describe two trends you found.

? Which reading looks wrong, and what could have caused it?

Your workbook: Lesson 1 · tick each step off as you go

QUICK CHECK

1. What does an accelerometer in a fitness tracker measure?

A Heart rate
B Blood oxygen
C Location
D Movement, used to count steps

2. What colour light do most optical heart-rate sensors use?

A Green
B Blue
C White
D Purple

3. What are wearables best at showing?

A Your exam results
B Trends over time
C Your thoughts
D Exact medical diagnoses

4. What does a continuous glucose monitor help with?

A Managing diabetes
B Counting steps
C Tracking location
D Measuring sleep

QUICK CHECK: ANSWERS

1. D: Movement, used to count steps

It senses motion.

2. A: Green

Blood absorbs green light well.

3. B: Trends over time

Single readings may be wrong.

4. A: Managing diabetes

It reads blood sugar levels.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how an optical heart-rate sensor measures heart rate.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe the sensors in a wearable and what they measure
- ✓ I can explain how an optical heart-rate sensor works
- ✓ I can evaluate the accuracy and usefulness of wearable data

LESSON 2 OF 6

AI IN MEDICINE

AI can analyse medical scans, spot patterns in health records and speed up admin, helping doctors work faster and catch disease earlier.

STARTER · 5 MINUTES

DO NOW

A doctor looks at 100 X-rays a day. How could AI help? What could go wrong?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how AI is used in medicine and explain why accuracy, bias and human oversight matter

SUCCESS CRITERIA: I CAN...

- ✓ I can describe uses of AI in healthcare
- ✓ I can explain false positives and false negatives
- ✓ I can explain why a human should stay in the loop

WHERE AI HELPS

- Medical imaging: spotting signs of cancer, fractures and eye disease in scans
- Triage: deciding who needs to be seen first
- Admin: AI that writes up notes so doctors can focus on patients
- Research: finding patterns in millions of records

KEY WORDS

MEDICAL IMAGING	Creating pictures of the inside of the body, such as X-rays and scans.
FALSE POSITIVE	A result saying a condition is present when it isn't.
FALSE NEGATIVE	A result missing a condition that is present.
BIAS	Unfairness in results, often because training data doesn't represent everyone.
HUMAN IN THE LOOP	A person who checks and is responsible for decisions an AI helps to make.

WHEN AI GETS IT WRONG

- False positive: AI says there's a problem when there isn't (worry, extra tests)
- False negative: AI misses a real problem (dangerous delay)
- Bias: if training data doesn't include everyone, AI may work worse for some groups
- AI can't explain its "thinking" easily

HUMANS IN THE LOOP

- AI as an assistant, not the final decision-maker
- Doctors check AI results and take responsibility
- AI tools must be tested and approved before use in the NHS
- Large trials, e.g. of AI in breast cancer screening, test whether AI really helps

ACTIVITY · 25 MINUTES · GROUPS OF 3

AI SCREENING SIMULATOR

1. Your teacher gives you 20 (made-up) "scan cards" and the AI's prediction for each, plus the real result.
2. Count the true positives, true negatives, false positives and false negatives.
3. Decide which type of error is more serious in cancer screening and why.
4. Write a rule for how doctors should use this AI.

In your workbook, answer:

? Which error is more serious in cancer screening, and why?

? Write your rule for how doctors should use the AI.

Your workbook: Lesson 2 · tick each step off as you go

QUICK CHECK

1. What is a false positive?

A A correct result
B A broken scanner
C A test says a problem is there when it isn't
D A test misses a real problem

2. What is a false negative?

A A positive attitude
B A correct result
C A test finds a problem that isn't there
D A test misses a real problem

3. Why might a medical AI be biased?

A It uses electricity
B Its training data didn't represent everyone
C It's too fast
D Doctors dislike it

4. What does human in the loop mean in medicine?

A AI makes all decisions
B Patients run the AI
C Nobody checks
D A doctor checks AI decisions and stays responsible

QUICK CHECK: ANSWERS

1. **C: A test says a problem is there when it isn't**
   Positive but wrong.

2. **D: A test misses a real problem**
   Negative but wrong.

3. **B: Its training data didn't represent everyone**
   Data shapes AI.

4. **D: A doctor checks AI decisions and stays responsible**
   Humans keep oversight.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe two ways AI is used in healthcare.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe uses of AI in healthcare
- ✓ I can explain false positives and false negatives
- ✓ I can explain why a human should stay in the loop

LESSON 3 OF 6

DNA IS DATA

DNA stores the instructions for life using four bases, A, C, G and T, in a sequence a bit like a computer code.

STARTER · 5 MINUTES

DO NOW

Computers store everything using two symbols: 0 and 1. How many symbols do you think DNA uses?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how DNA stores information as a code, and how genome sequencing turns it into data

SUCCESS CRITERIA: I CAN...

- ✓ I can explain how DNA is a code using four letters
- ✓ I can describe genome sequencing and why it's become possible on a large scale
- ✓ I can compare storing data in DNA with storing it in binary

THE CODE OF LIFE

- DNA is in almost every cell of your body
- Its code uses four bases: A, C, G, T
- A gene is a section of DNA that gives instructions, e.g. to make a protein
- Your genome is your complete set of DNA: about 3 billion letters

KEY WORDS

DNA	The molecule that carries the instructions for living things, coded with four bases: A, C, G and T.
GENOME	The complete set of DNA in an organism.
GENE	A section of DNA that gives an instruction, often to make a protein.
SEQUENCING	Reading the order of the bases in DNA.
BIOINFORMATICS	Using computers to store and analyse biological data.

READING THE CODE

- Sequencing reads the order of the letters
- Human Genome Project: 13 years and billions of dollars, finished 2003
- Today: a genome can be sequenced in hours for a few hundred pounds
- Bioinformatics: using computers to analyse biological data

GENOMICS IN THE NHS

- Diagnosing rare diseases by finding changes in a child's genome
- Choosing cancer treatments based on the tumour's DNA
- The Generation Study is sequencing the genomes of newborn babies
- Huge storage and computing power needed: each genome is many gigabytes

ACTIVITY · 20 MINUTES · PAIRS

DNA DATA LAB

1. Use the code 00 = A, 01 = C, 10 = G, 11 = T.
2. Encode your initials: convert each letter to 8-bit ASCII, then to DNA letters.
3. Swap with a partner and decode their DNA message.
4. Answer: why can DNA store more data in the same space than binary?

In your workbook, answer:

? Encode the letter "A" (ASCII 65 = 01000001) as DNA.

? Why can each DNA letter store more information than a single bit?

Your workbook: Lesson 3 · tick each step off as you go

QUICK CHECK

1. How many different bases (letters) does DNA use?

A 4
B 26
C 10
D 2

2. What is a genome?

A A type of cell
B A single gene
C An organism's complete set of DNA
D A computer virus

3. What is sequencing?

A Writing a story
B Reading the order of letters in DNA
C Sorting files
D Counting steps

4. What is bioinformatics?

A Using computers to analyse biological data
B A fitness app
C Biology without computers
D A type of DNA

QUICK CHECK: ANSWERS

1. A: 4

A, C, G and T.

2. C: An organism's complete set of DNA

About 3 billion letters in humans.

3. B: Reading the order of letters in DNA

It turns DNA into data.

4. A: Using computers to analyse biological data

Computing meets biology.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how DNA is similar to computer code.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain how DNA is a code using four letters
- ✓ I can describe genome sequencing and why it's become possible on a large scale
- ✓ I can compare storing data in DNA with storing it in binary

LESSON 4 OF 6

ENGINEERING BIOLOGY

Engineering biology treats living systems like something that can be designed, built and tested.

STARTER · 5 MINUTES

DO NOW

If you could change one thing about a plant to help the planet, what would it be?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe engineering biology, gene editing and AI protein prediction, and discuss the ethical questions they raise

SUCCESS CRITERIA: I CAN...

- ✓ I can explain what engineering biology is
- ✓ I can describe how CRISPR gene editing works in simple terms
- ✓ I can explain how AI is speeding up biology, and discuss ethical issues

BIOLOGY AS ENGINEERING

- Engineering biology: designing and building with living systems
- Cells "programmed" to make medicines, materials or fuels
- Insulin for diabetes is made by engineered microbes
- The UK sees engineering biology as a key future technology

KEY WORDS

ENGINEERING BIOLOGY	Designing, building and testing living systems to solve problems.
CRISPR	A gene-editing tool that finds and cuts a specific DNA sequence.
PROTEIN	A molecule that does a job in living things; its shape decides its job.
ALPHAFOLD	An AI system from Google DeepMind that predicts the 3D shape of proteins.
DESIGN-BUILD-TEST-LEARN	The cycle of designing, making, testing and improving something.

CRISPR: FIND AND EDIT

- CRISPR is a tool that finds a specific DNA sequence and cuts it
- Like "find and replace" for DNA
- 2023: the UK was the first country to approve a CRISPR-based treatment, for sickle cell disease
- Changes to embryos that could be passed on are not allowed in the UK

AI MEETS BIOLOGY

- Proteins do most jobs in our cells; their 3D shape decides what they do
- AlphaFold, from Google DeepMind in London, predicts protein shapes from their code
- Its creators shared the 2024 Nobel Prize in Chemistry
- Faster drug discovery, new enzymes to break down plastic

ACTIVITY · 25 MINUTES · GROUPS OF 3

BIO-DESIGNERS

1. Choose a problem: plastic pollution, a crop disease, or a rare genetic illness.
2. Design an engineering-biology solution using the design-build-test-learn cycle.
3. Explain where computing or AI would help.
4. Complete an ethics checklist: benefits, risks, who might be left out.

In your workbook, answer:

? Describe your solution and the role of computing.

? What is the biggest ethical risk of your idea, and how would you manage it?

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. What is engineering biology?

A Designing, building and testing with living systems
B Studying rocks
C Building bridges
D Repairing computers

2. What does CRISPR do?

A Sequences a whole genome
B Counts steps
C Prints proteins
D Finds and cuts a specific DNA sequence

3. What does AlphaFold predict?

A The 3D shape of proteins
B Exam results
C Football scores
D The weather

4. Which disease could be treated with the first UK-approved CRISPR treatment?

A Toothache
B Flu
C Sickle cell disease
D Chickenpox

QUICK CHECK: ANSWERS

1. **A: Designing, building and testing with living systems**
   Engineering ideas applied to biology.

2. **D: Finds and cuts a specific DNA sequence**
   Like "find and replace" for DNA.

3. **A: The 3D shape of proteins**
   Shape decides what a protein does.

4. **C: Sickle cell disease**
   Approved in 2023.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how CRISPR edits DNA.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain what engineering biology is
- ✓ I can describe how CRISPR gene editing works in simple terms
- ✓ I can explain how AI is speeding up biology, and discuss ethical issues

LESSON 5 OF 6

YOUR HEALTH DATA

Health and genetic data are among the most sensitive information about a person.

STARTER · 5 MINUTES

DO NOW

Who do you think should be allowed to see your medical records: you, your doctor, a researcher, an insurance company, a tech company?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain why health data is sensitive, how it is protected by law, and the benefits and risks of sharing it

SUCCESS CRITERIA: I CAN...

- ✓ I can explain why health and genetic data are sensitive
- ✓ I can describe how UK law protects health data
- ✓ I can weigh up the benefits and risks of sharing health data

WHY HEALTH DATA IS SPECIAL

- It's deeply personal and can't be changed like a password
- Genetic data also reveals things about your relatives
- It could be misused for discrimination, e.g. in jobs or insurance
- Criminals target health data because it's valuable

KEY WORDS

SPECIAL CATEGORY DATA	Highly sensitive personal data, such as health and genetic data, with extra legal protection.
CONSENT	Clear, informed agreement.
ANONYMISATION	Removing details that identify people from data.
DATA BREACH	When personal data is lost, stolen or seen by people who shouldn't see it.

THE LAW PROTECTS IT

- UK GDPR: health and genetic data are "special category data"
- Organisations need a strong legal reason and extra security
- Consent: agreeing, knowing what you're agreeing to
- NHS patients can choose to opt out of their data being used for research and planning

SHARING FOR GOOD

- Anonymised data can help find cures and plan services
- Health apps and wearables may share data with companies
- Check privacy settings and what an app does with your data
- Trust depends on transparency, security and benefit to patients

ACTIVITY · 25 MINUTES · GROUPS OF 4

THE DATA DILEMMA

1. Your group sits on a (fictional) hospital ethics committee.
2. Read three requests: a university wants anonymised scan data to train cancer AI; a fitness app wants to sell users' heart data to advertisers; an insurer wants customers' genetic test results.
3. Decide whether to approve, approve with conditions, or refuse each one.
4. Justify each decision using the law, consent and benefit to patients.

In your workbook, answer:

? Give your decisions for the three requests.

? What conditions would make sharing data acceptable?

Your workbook: Lesson 5 · tick each step off as you go

QUICK CHECK

1. How does UK GDPR treat health and genetic data?

A As unimportant
B As special category data needing extra protection
C As data anyone can sell
D As public data

2. What is consent?

A A type of encryption
B Saying nothing
C Clear, informed agreement
D Clicking without reading

3. What does anonymisation do?

A Encrypts the whole internet
B Shares names publicly
C Removes details that identify people
D Deletes all data

4. Why is genetic data especially sensitive?

A It's very small
B It's always public
C It's boring
D It can't be changed and reveals information about relatives

QUICK CHECK: ANSWERS

1. **B: As special category data needing extra protection**
It's especially sensitive.

2. **C: Clear, informed agreement**
You must know what you're agreeing to.

3. **C: Removes details that identify people**
It protects identity.

4. **D: It can't be changed and reveals information about relatives**
DNA is shared with family.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why health data needs extra protection.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain why health and genetic data are sensitive
- ✓ I can describe how UK law protects health data
- ✓ I can weigh up the benefits and risks of sharing health data

LESSON 6 OF 6

DESIGN A HEALTH-TECH SOLUTION

Good health tech solves a real health problem for real people.

STARTER · 5 MINUTES

DO NOW

Think of someone you know who has to manage a health condition every day. What makes it harder for them?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Design a health-tech solution that is user-centred, accessible, accurate and safe

SUCCESS CRITERIA: I CAN...

- ✓ I can identify a health problem and the people affected
- ✓ I can design a health-tech solution with accessibility and privacy built in
- ✓ I can explain how health tech is tested and regulated

START WITH PEOPLE

- Who has the problem: patients, carers, clinicians?
- Involve them from the start (co-design)
- Will it work for older people, children, people with disabilities?
- Does it reduce or increase inequality?

KEY WORDS

USER-CENTRED DESIGN	Designing around the needs of the people who will use something.
ACCESSIBILITY	Making technology usable by as many people as possible, including disabled people.
CLINICAL TRIAL	A careful study that tests a treatment or device with volunteers.
REGULATION	Rules set and enforced by an official body.

DESIGN IT SAFELY

- Accuracy: test against the real result, with diverse users
- Privacy by design: collect only what's needed, keep it secure
- Accessibility: large text, voice control, simple language
- What happens if it fails, or the battery dies?

TESTING AND REGULATION

- Health tech that diagnoses or treats is a medical device
- In the UK the MHRA regulates medical devices and medicines
- Clinical trials test safety and whether it really works
- Ongoing monitoring after launch

ACTIVITY · 35 MINUTES · GROUPS OF 3

HEALTH-TECH PITCH

1. Choose a user: a teenager with asthma, a grandparent living alone, or a young person with type 1 diabetes.
2. Identify their biggest daily challenge and design a health-tech solution (an app, wearable or smart device).
3. Complete a design card covering accessibility, accuracy, privacy and safety.
4. Pitch your idea in 2 minutes, explaining how it would be tested.

In your workbook, answer:

? Describe your user, their challenge and your solution.

? How would you make sure it's accurate, accessible and private?

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. What should health-tech design start with?

A The price
B The logo
C The newest gadget
D The needs of patients, carers and clinicians

2. Which UK body regulates medical devices?

A Ofcom
B The MHRA
C The CAA
D The DVLA

3. What is a clinical trial?

A A computer game
B A fitness challenge
C A careful study testing a treatment or device with volunteers
D A court case

4. Which is an accessibility feature?

A Only one colour
B Large clear text and voice control
C Tiny text
D Complicated menus

QUICK CHECK: ANSWERS

1. **D: The needs of patients, carers and clinicians**
   User-centred design.

2. **B: The MHRA**
   Medicines and Healthcare products Regulatory Agency.

3. **C: A careful study testing a treatment or device with volunteers**
   It tests safety and effectiveness.

4. **B: Large clear text and voice control**
   Helps more people use it.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why health-tech products must be tested carefully before use.

Look back at today's success criteria: can you tick them all?

- ✓ I can identify a health problem and the people affected
- ✓ I can design a health-tech solution with accessibility and privacy built in
- ✓ I can explain how health tech is tested and regulated

END OF TOPIC

ENGINEERING BIOLOGY AND HEALTH TECH

Revise each lesson in your workbook: the notes, key words and quick checks. Then take the topic test when Mr Butler sets it.